

Angles, Polygons & Parallel Lines

Quick reference sheet · GCSE Maths, Geometry & Measures · keep handy for revision

1. Angle facts: lines and points

Angles on a straight line sum to **180°**. Angles around a point sum to **360°**. Vertically opposite angles (where two lines cross) are **equal**.

2. Triangles & quadrilaterals

Angles in a **triangle** sum to **180°**. Angles in a **quadrilateral** sum to **360°** (split into two triangles). Isosceles triangles have two equal angles; equilateral triangles have three 60° angles.

3. Angles in parallel lines

Angle pair	Shape	Rule
Corresponding	F-shape	Equal
Alternate	Z-shape	Equal
Co-interior (allied)	C/U-shape	Sum to 180°
Vertically opposite	X-shape	Equal

Corresponding and alternate angles are equal; co-interior (allied) angles add up to 180°.

4. Interior & exterior angles of polygons

Sum of interior angles = **$(n - 2) \times 180^\circ$** , where n is the number of sides. Exterior angles of any convex polygon always sum to **360°**. At each vertex, interior + exterior = 180°.

Polygon	Sides (n)	Interior angle sum
Triangle	3	180°
Quadrilateral	4	360°
Pentagon	5	540°
Hexagon	6	720°
Heptagon	7	900°
Octagon	8	1080°

5. Regular polygons

Each exterior angle = **$360^\circ \div n$** . Each interior angle = **$180^\circ - \text{exterior angle}$** . Working backwards: $n = 360^\circ \div \text{exterior angle}$.

Regular polygon	n	Exterior	Interior
Square	4	90°	90°
Pentagon	5	72°	108°
Hexagon	6	60°	120°
Octagon	8	45°	135°
Decagon	10	36°	144°

6. Common mistakes

Using 360° instead of 180° for a triangle, or vice versa for a quadrilateral.

Mixing up alternate (equal) and co-interior (sum to 180°) angles — check whether the angles are on the same or opposite sides of the transversal.

Forgetting that exterior angles of ANY polygon sum to 360°, regardless of the number of sides.

Using the interior angle formula $(n-2) \times 180^\circ$ when the question actually gives an exterior angle, or vice versa.